TEAM NEXORA · DOSM DATATHON 2026

HOW DOES GEOPOLITICAL RISK AFFECT TOURIST ARRIVALS TO MALAYSIA?

Official data and machine learning on Malaysia's top 20 source markets

Chin William · Tang Xin Yi ·
Janis Ong Jing Yao · Tee Xin Yee
UTAR · Advisor: Dr. Kelwin Tan Seen Tiong



THE CONTEXT

Tourism matters to Malaysia

15.9% of GDP

3.7M tourism jobs

hotels · airlines · homestays

+0.08

risk vs arrivals, all markets combined

THE PROBLEM

Nationally, risk seems not to matter.

THE GAP

- Lee et al. (2021): no national link
- Nobody had checked each country separately



OUR OBJECTIVE

Test every market, then forecast ahead



- Test each of our top 20 markets
- Forecast arrivals 1–4 months ahead

When risk goes up, do tourists come…

A less

B more

C both

HOW WE DID IT

Four steps, one rule



Collect	Date	Model	Test
69,040 daily rows → 2,300 country-months	Publish day, not event day	Ridge for the link, LightGBM machine learning to forecast	5 rolling tests + COVID stress test → Power BI

Only learns from numbers already published

FIRST FINDING

The answer is both



C · Both

16
fewer tourists
e.g. Nepal, South Korea, Germany

3
more tourists
Singapore, Indonesia, China

1
almost no change
Thailand

20 modelled markets · associations, not causes

KEY INSIGHT

Drops and rises cancel out



16 markets + 3 markets = +0.08 national

Watch risk market by market

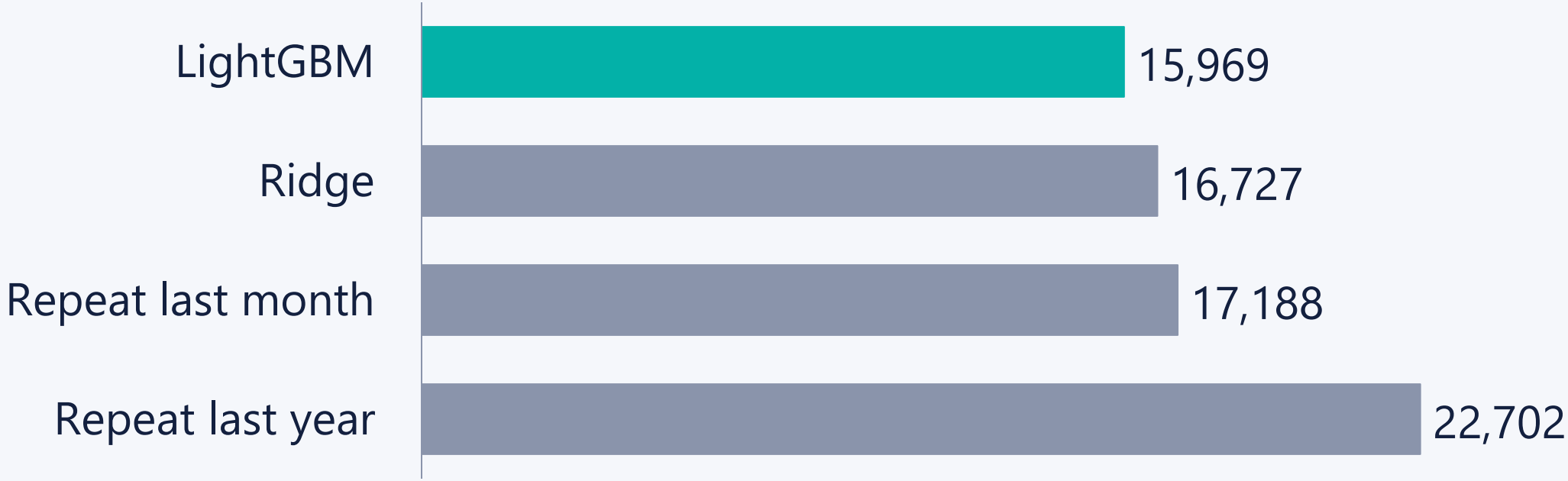
Patterns, not proof of cause

SECOND FINDING

LightGBM had the smallest error



Average error · arrivals per market per month · tested on past data, as if live



HOW FAR AHEAD

Most accurate 1–2 months ahead



LightGBM average error, by months ahead

Month 1	Month 2	Month 3	Month 4
11,817	9,571	21,504	20,983

under 12,000

about double

⚠ During COVID, repeating last month won

POWER BI DASHBOARD

Live demo: four questions

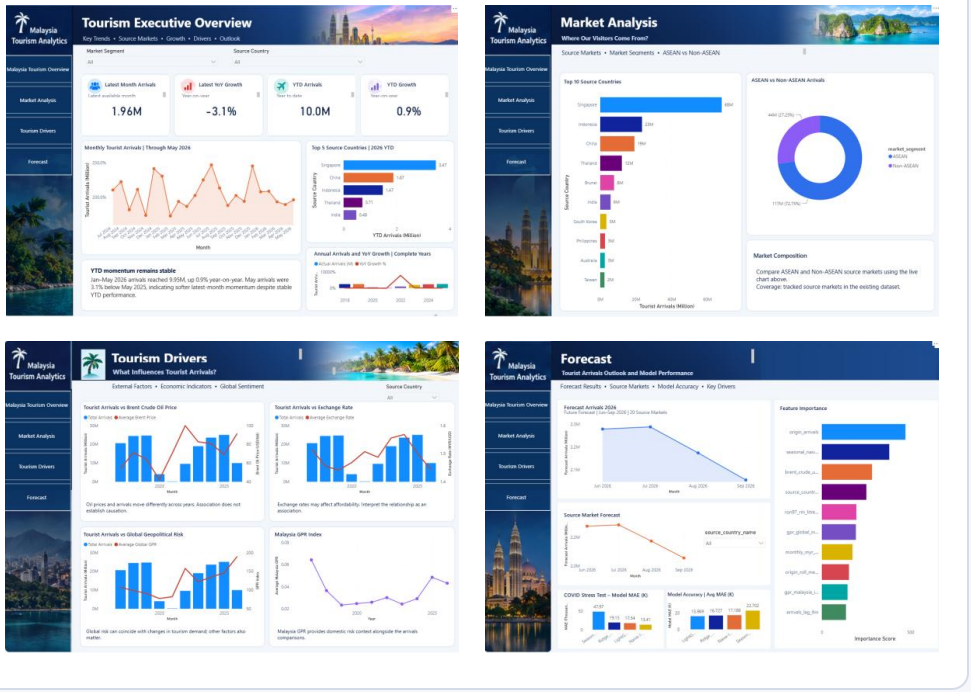


Start: Overview | **20 modelled markets**

1. Which markets matter? — **Market Analysis**
2. What moves arrivals? — **Tourism Drivers**
3. What do we forecast? — **Forecast**
4. How accurate is it? — **Forecast**

Demo backup ▸

● LIVE



WHY IT MATTERS

Useful, on-theme and realistic

1 Useful

Tourism Malaysia

DOSM

Hotels, airlines, homestays

2 On-theme

SDG 8.9

Supports SDG 12.b monitoring

National Tourism Policy 2020–2030

3 Realistic

Open official data

One-command monthly update

More markets and states

⚠ Possible uses: impact not yet measured



CONCLUSION

Three things we found



1

Each market reacts differently

2

Market by market shows what one number hides

3

Best forecasts 1–2 months ahead

OUR HONEST LIMITS

- Patterns, not causes
- Shocks like COVID can't be predicted
- 20 markets only; forecast ranges next

THE ANSWER

FOR 16 OF OUR 20 MARKETS, HIGHER RISK GOES WITH FEWER TOURISTS, BUT NOT FOR ALL

That's why we look market by market.

Terima kasih.

Q&A Accuracy Markets Data & design Demo



github.com/william20808/nexora_dosm

BACKUP A · ACCURACY

Best at 1–2 months, robust to the lag, humbled by COVID



MAE BY HORIZON · ARRIVALS PER MARKET-MONTH

Horizon	Naive-last	Seasonal-naive	Ridge	LightGBM
	repeat last month	repeat last year		
Month 1	12,752	23,802	12,117	**11,817**
Month 2	9,671	20,404	9,757	**9,571**
Month 3	25,157	**21,191**	24,258	21,504
Month 4	21,171	25,413	**20,775**	20,983
Aggregate	17,188	22,702	16,727	**15,969**

5 primary folds, Mar 2023–Sep 2025 · best per row highlighted

Ridge won 7 of 20 test cells

COVID STRESS · 2020–22

Horizon	Naive-last	LightGBM
Month 1	**9,647**	12,805
Month 2	**12,366**	16,279
Month 3	**15,511**	20,730
Month 4	**16,134**	20,334

Naive-last wins every horizon

LAG TEST · SAME MODEL CHOICE

Horizon	Lag 3	Lag 4 (used)
Month 1	12,611	11,817
Month 2	10,762	9,571
Month 3	20,835	21,504
Month 4	20,509	20,983

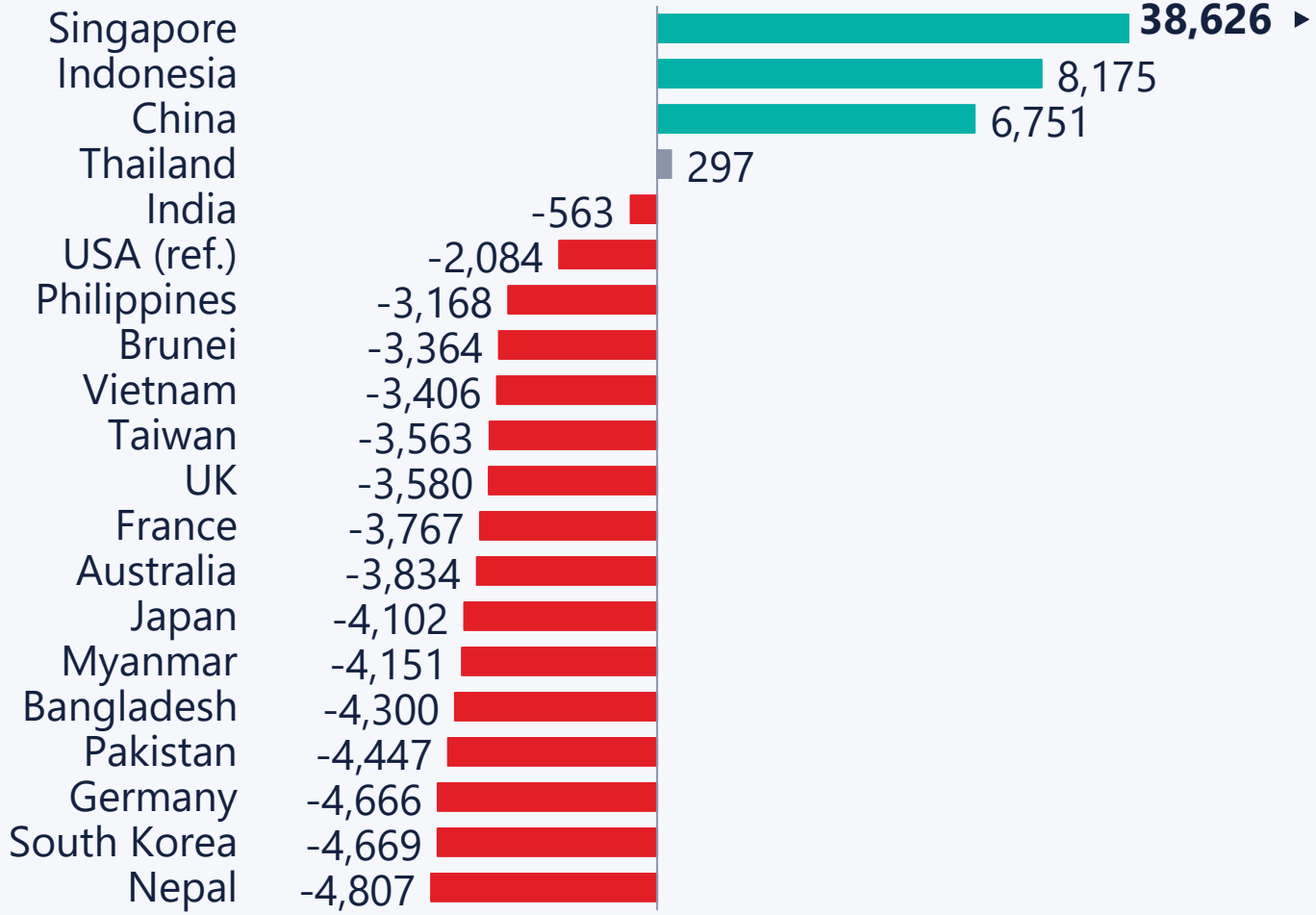
◂ Q&A index

BACKUP B · MARKETS

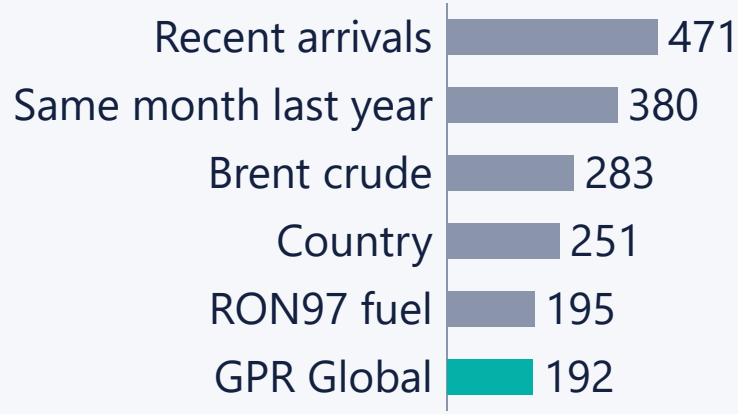
16 of 20 markets slope down; the forecast leans on past arrivals



Arrivals per +1 SD of GPR Global · Ridge, USA as reference



LIGHTGBM SPLIT COUNT · TOP 6



READ WITH CARE

Raw correlation +0.08
GPR vs time +0.54
Associations, not causes
SHAP is in-sample

◂ Q&A index

BACKUP C · DATA & DESIGN

Every input is dated by release; every design choice has a reason



Source	Inputs	Available after
Tourism Malaysia & Immigration	arrivals by nationality	assumed 4 months (3 tested)
Caldara & Iacoviello	GPR Global, Malaysia	≈1 month
World Bank	Brent crude	≈1 month
Bank Negara Malaysia	MYR exchange rates (+ BDT cross-rate)	real time
Ministry of Finance	RON97, diesel	weekly
DOSM MEI	leading & coincident indices	≈2 months → excluded

Why pooled?
~111 months per market is too few

Why direct multi-horizon?
No error carried month to month

Why MAE and RMSE, not MAPE?
Tiny markets blow up percentages

Why USA as reference?
Identifies every market's slope

Why drop DOSM MEI?
3–4 months stale; near-zero weight

Why LightGBM over Ridge?
Lowest total error, pooled design

SDG OUTCOME INDICATORS

- SDG 8.9: tourism jobs, wages, local SME participation
- SDG 12.b: energy, water, waste, traffic

◂ Q&A index

69,040 daily rows → 2,300 country-months · same rule in backtest and deployment

BACKUP D · DEMO

Demo backup: the same four questions, as screenshots

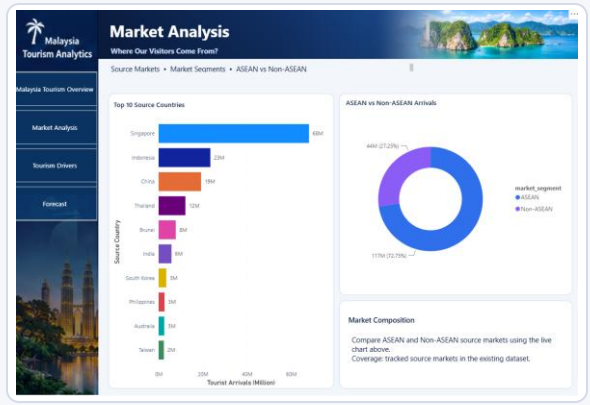




Start

Overview

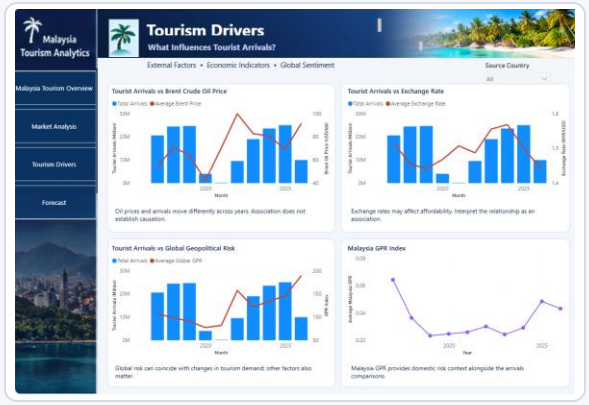
20 modelled markets



1

Which markets matter?

Market Analysis



2

What moves arrivals?

Tourism Drivers



3 & 4

What do we forecast? How accurate is it?

Forecast

◂ Q&A index

◂ Back to demo